StudySapiens Board MCQs

Chapter 9 — Human Digestive System

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. The gastrointestinal tract extends from:

(a) Stomach to anus only
(b) Oesophagus to rectum only
(c) Mouth to anus
(d) Mouth to stomach only

2. Which of the following is an accessory digestive organ?

(a) Small intestine
(b) Stomach
(c) Oesophagus
(d) Liver

3. Mechanical digestion in the oral cavity is called:

(a) Segmentation
(b) Emulsification
(c) Peristalsis
(d) Mastication

4. Salivary amylase begins the digestion of:

(a) Starch
(b) Fat
(c) DNA
(d) Protein

5. Which of the following is NOT a salivary gland?

(a) Submaxillary gland
(b) Sublingual gland
(c) Parotid gland
(d) Adrenal gland

6. Saliva contains thiocyanate ions that:

(a) Digest proteins
(b) Neutralize bile
(c) Kill microorganisms
(d) Emulsify fats

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7. The moist mass of chewed food mixed with saliva is called:

(a) Bolus
(b) Chyme
(c) Faeces
(d) Chyle

8. During swallowing, the trachea is closed by the:

(a) Tongue only
(b) Uvula only
(c) Soft palate only
(d) Epiglottis

9. The common passageway for food and air is the:

(a) Trachea
(b) Oesophagus
(c) Larynx
(d) Pharynx

10. Food moves through the oesophagus by:

(a) Peristalsis
(b) Segmentation only
(c) Diffusion
(d) Osmosis

11. The sphincter at the lower end of the oesophagus is the:

(a) Pyloric sphincter
(b) Cardiac sphincter
(c) Ileocaecal valve
(d) Anal sphincter

12. The stomach is described as:

(a) Straight
(b) J-shaped only
(c) Coiled like a spring
(d) U-shaped

13. Which of the following is NOT a layer of the stomach wall?

(a) Serosa
(b) Epidermis
(c) Mucosa
(d) Submucosa

14. Parietal cells secrete:

(a) Pepsinogen
(b) Bile
(c) Mucus only
(d) HCl

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15. Chief cells secrete:

(a) Insulin
(b) Pepsinogen
(c) Bile
(d) HCl

16. HCl in the stomach activates:

(a) Pepsinogen to pepsin
(b) Lipase to bile
(c) Amylase to maltase
(d) Trypsinogen to trypsin

17. The pH of stomach contents is approximately:

(a) 2–3
(b) 5–6
(c) 9–10
(d) 7–8

18. Pepsin digests proteins into:

(a) Fatty acids
(b) Amino acids only
(c) Polypeptides and peptides
(d) Glucose

19. The semi-liquid acidic mass leaving the stomach is called:

(a) Bolus
(b) Chyle
(c) Chyme
(d) Bile

20. The hormone gastrin stimulates:

(a) Insulin release
(b) Bile release only
(c) Gastric juice secretion
(d) Saliva only

21. The hormone that stops HCl release when food enters the small intestine is:

(a) Secretin only
(b) CCK only
(c) Somatostatin
(d) Gastrin

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22. The pyloric sphincter controls movement of chyme into the:

(a) Colon only
(b) Oesophagus
(c) Ileum only
(d) Duodenum

23. The longest part of the alimentary canal is the:

(a) Oesophagus
(b) Large intestine
(c) Stomach
(d) Small intestine

24. The three parts of the small intestine are:

(a) Duodenum, jejunum, ileum
(b) Ascending, transverse, descending
(c) Cardiac, fundus, pyloric
(d) Caecum, colon, rectum

25. Pancreatic juice is:

(a) Strongly acidic
(b) Neutral only
(c) Alkaline
(d) Bile-free always

26. The bicarbonate in pancreatic juice mainly:

(a) Activates pepsin
(b) Digests protein
(c) Emulsifies fat
(d) Neutralizes acidic chyme

27. Trypsinogen is activated by:

(a) Enterokinase
(b) Amylase
(c) HCl
(d) Bile

28. Pancreatic lipase digests lipids into:

(a) Nucleotides
(b) Amino acids
(c) Glycerol and fatty acids
(d) Glucose

29. Bile is produced by the:

(a) Liver
(b) Stomach
(c) Gallbladder
(d) Pancreas

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30. Bile helps fat digestion by:

(a) Emulsifying fats
(b) Hydrolysing proteins
(c) Neutralizing acid only
(d) Digesting starch

31. Bile contains:

(a) No digestive enzymes
(b) Amylase only
(c) Pepsin only
(d) Lipase only

32. The gallbladder stores and concentrates:

(a) Bile
(b) Pancreatic juice
(c) Gastric juice
(d) Saliva

33. The hormone secretin is released in response to:

(a) Glucose in blood only
(b) Protein in the stomach only
(c) Acidic chyme in the duodenum
(d) Fat in the mouth

34. Cholecystokinin (CCK) is released in response to:

(a) Acid only
(b) Partially digested proteins and fats
(c) Glucose only
(d) Water only

35. CCK stimulates the gallbladder to release:

(a) Insulin
(b) Saliva
(c) Bile
(d) Gastrin

36. Villi are found mainly in the:

(a) Large intestine
(b) Small intestine
(c) Oesophagus
(d) Stomach

37. Microvilli are found on:

(a) Muscle fibres only
(b) Epithelial cells of villi
(c) Neurons
(d) Red blood cells

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38. The lymphatic vessel inside a villus is called a:

(a) Capillary only
(b) Lacteal
(c) Arteriole only
(d) Venule only

39. Glucose and amino acids are absorbed into the:

(a) Lacteals only
(b) Blood capillaries
(c) Stomach glands
(d) Bile ducts

40. Fatty acids and glycerol are reassembled into triglycerides and packaged as:

(a) Chylomicrons
(b) Bolus
(c) Chyme
(d) Glycogen

41. The hepatic portal vein carries absorbed nutrients to the:

(a) Kidneys directly
(b) Liver
(c) Heart directly
(d) Lungs directly

42. The large intestine mainly absorbs:

(a) Proteins
(b) Fats only
(c) Glucose only
(d) Water, salts and vitamin K

43. Vitamin K produced by intestinal bacteria is important for:

(a) Bone contraction
(b) Oxygen transport
(c) Blood coagulation
(d) Vision only

44. The vermiform appendix is attached to the:

(a) Duodenum
(b) Stomach
(c) Caecum
(d) Rectum

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45. In humans, the appendix is considered:

(a) A major enzyme source
(b) Essential for digestion
(c) Vestigial
(d) A bile store

46. The colon is divided into:

(a) Ascending, transverse and descending parts
(b) Duodenum, jejunum, ileum
(c) Cardiac, fundus, pyloric
(d) Cortex and medulla

47. Faeces are temporarily stored in the:

(a) Rectum
(b) Caecum only
(c) Stomach
(d) Ileum

48. The internal anal sphincter is made of:

(a) Skeletal muscle
(b) Smooth muscle
(c) Cardiac muscle
(d) Cartilage

49. The external anal sphincter is made of:

(a) Skeletal muscle
(b) Elastic cartilage
(c) Smooth muscle
(d) Cardiac muscle

50. In infants, egestion is:

(a) Controlled by the liver
(b) Absent
(c) An involuntary reflex
(d) Fully voluntary

51. Segmentation in the intestine mainly:

(a) Absorbs water only
(b) Mixes food with secretions
(c) Propels food only
(d) Produces bile

52. Antiperistalsis can lead to:

(a) Constipation only
(b) Bile release
(c) Diarrhoea only
(d) Vomiting

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53. The pancreas is:

(a) Both exocrine and endocrine
(b) Only endocrine
(c) Neither
(d) Only exocrine

54. The endocrine pancreas secretes:

(a) Insulin and glucagon
(b) Pancreatic juice only
(c) Bile only
(d) Gastrin only

55. The liver converts toxic ammonia into:

(a) Bile only
(b) Glycogen only
(c) Urea
(d) Glucose

56. The liver stores excess glucose as:

(a) Starch
(b) Cellulose
(c) Fat only
(d) Glycogen

57. Jaundice may occur when bile pigments accumulate in the:

(a) Sweat only
(b) Saliva only
(c) Blood
(d) Urine only

58. Gallstones may form from precipitation of:

(a) Cholesterol
(b) Urea
(c) Amino acids
(d) Glucose

59. The enzyme erepsin acts on:

(a) Starch
(b) DNA
(c) Lipids
(d) Dipeptides

60. Maltase converts maltose into:

(a) Galactose only
(b) Fructose only
(c) Glucose
(d) Sucrose

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61. Sucrase converts sucrose into:

(a) Glucose and galactose
(b) Glucose and fructose
(c) Two glucose molecules
(d) Amino acids

62. Lactase converts lactose into:

(a) Glucose and fructose
(b) Two glucose molecules
(c) Fatty acids
(d) Glucose and galactose

63. The alkaline emulsion formed after intestinal digestion is called:

(a) Chyle
(b) Bile
(c) Chyme
(d) Bolus

64. Which of the following is NOT a function of the liver?

(a) Production of insulin
(b) Detoxification
(c) Glycogen storage
(d) Bile production

65. The muscularis externa of the stomach includes:

(a) Longitudinal, circular and oblique muscles
(b) Only smooth circular muscle
(c) Only longitudinal muscle
(d) Only skeletal muscle

66. The stomach mucosa is protected from acid by:

(a) Pancreatic juice only
(b) Mucus
(c) Bile only
(d) Saliva only

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ANSWER KEY WITH BRIEF REASONS

1. (c) Mouth to anus — The GI tract is a continuous tube.

2. (d) Liver — Accessory organs aid digestion but food does not pass through them.

3. (d) Mastication — Chewing breaks food into smaller pieces.

4. (a) Starch — It hydrolyses polysaccharides to maltose.

5. (d) Adrenal gland — Adrenal is an endocrine gland.

6. (c) Kill microorganisms — Thiocyanate has antimicrobial action.

7. (a) Bolus — Bolus is formed in the mouth.

8. (d) Epiglottis — Epiglottis prevents food entry into airway.

9. (d) Pharynx — Pharynx serves both systems.

10. (a) Peristalsis — Wave-like contractions propel food.

11. (b) Cardiac sphincter — It controls entry into the stomach.

12. (d) U-shaped — The textbook describes it as U-shaped.

13. (b) Epidermis — Epidermis is skin, not stomach wall.

14. (d) HCl — Oxyntic/parietal cells produce acid.

15. (b) Pepsinogen — Chief cells produce the inactive protease.

16. (a) Pepsinogen to pepsin — Acid converts pepsinogen to active pepsin.

17. (a) 2–3 — Gastric acid makes the lumen strongly acidic.

18. (c) Polypeptides and peptides — Pepsin begins protein digestion.

19. (c) Chyme — Stomach mixing forms chyme.

20. (c) Gastric juice secretion — Gastrin promotes gastric secretion.

21. (c) Somatostatin — Somatostatin inhibits acid secretion.

22. (d) Duodenum — It regulates gastric emptying.

23. (d) Small intestine — The small intestine is about 6 m long.

34. (b) Partially digested proteins and fats — CCK responds to nutrients in chyme.

35. (c) Bile — CCK promotes bile release.

36. (b) Small intestine — Villi increase absorptive surface.

37. (b) Epithelial cells of villi — They form the brush border.

38. (b) Lacteal — Lacteals absorb fat products.

39. (b) Blood capillaries — They enter the hepatic portal circulation.

40. (a) Chylomicrons — Chylomicrons enter lacteals.

41. (b) Liver — Liver processes nutrients first.

42. (d) Water, salts and vitamin K — It concentrates faeces.

43. (c) Blood coagulation — Vitamin K is needed for clotting.

44. (c) Caecum — Appendix arises from the caecum.

45. (c) Vestigial — It has little digestive function.

46. (a) Ascending, transverse and descending parts — These are colonic regions.

47. (a) Rectum — Rectum stores waste before egestion.

48. (b) Smooth muscle — It is involuntary.

49. (a) Skeletal muscle — It is under voluntary control.

50. (c) An involuntary reflex — Higher brain control is immature.

51. (b) Mixes food with secretions — It increases contact with enzymes.

52. (d) Vomiting — Reverse movement expels contents upward.

53. (a) Both exocrine and endocrine — It secretes enzymes and hormones.

54. (a) Insulin and glucagon — Islets release hormones into blood.

55. (c) Urea — Urea is excreted by kidneys.

56. (d) Glycogen — Glycogen is animal storage carbohydrate.

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24. (a) Duodenum, jejunum, ileum — These are the small-intestinal regions.

25. (c) Alkaline — Bicarbonate makes it alkaline.

26. (d) Neutralizes acidic chyme — It raises duodenal pH.

27. (a) Enterokinase — Enterokinase from duodenum activates trypsinogen.

28. (c) Glycerol and fatty acids — Lipase hydrolyses fats.

29. (a) Liver — Liver makes bile; gallbladder stores it.

30. (a) Emulsifying fats — Bile salts increase fat surface area.

31. (a) No digestive enzymes — Bile acts physically, not enzymatically.

32. (a) Bile — It releases bile into the duodenum.

33. (c) Acidic chyme in the duodenum — Secretin responds to acidity.

57. (c) Blood — Impaired bile flow causes yellowing.

58. (a) Cholesterol — Cholesterol can crystallize in bile.

59. (d) Dipeptides — Erepsin completes peptide digestion.

60. (c) Glucose — Maltase hydrolyses maltose to glucose.

61. (b) Glucose and fructose — Sucrase splits table sugar.

62. (d) Glucose and galactose — Lactase digests milk sugar.

63. (a) Chyle — Chyle is the intestinal product.

64. (a) Production of insulin — Insulin is pancreatic.

65. (a) Longitudinal, circular and oblique muscles — Three layers aid churning.

66. (b) Mucus — Mucus forms a protective barrier.

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